

PRANJAY DHAWAN

Chandigarh

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My Portfolio

Education

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Electronics and Communication Engineering

August 2023 – July 2026

Patiala, Punjab

Chandigarh College of Engineering and Technology

Diploma in Engineering in Electronics and Communication Engineering

August 2020 – June 2023

Chandigarh

Experience

Zipbolt Innovation Pvt Ltd

Product R&D Engineer - Internship

January 2026 – July 2026

Chandigarh

- Developed hardware architecture and PCB for the ZipSure AI EV monitoring system for real-time battery analytics consisting of sensory node and gateway node.
- Designed and prototyped EV-compatible enclosures for ZipSure AI using SolidWorks and in-house 3D printing.
- Enhanced firmware for ESP32-based IoT devices at charging station by adding advanced debugging and diagnostics features.

The MindCrafters

Core Electronics Engineer - Internship

February 2022 – July 2023

Chandigarh

- PCB Design and Electronics: Engineered and fabricated 4 custom PCBs using EasyEDA for robotic applications, ensuring seamless integration of electronic components.
- IoT and Robotics Projects: Spearheaded the development of 4 humanoid robot prototypes and 3 IoT systems utilizing ThingSpeak, improving project performance and data analytics capabilities.
- 3D Printing and CAD Design: Executed 3D printing and CAD design for 50+ complex models using SketchUp, contributing to projects development.

Projects

Drone - Garud | Altium, Pixhawk, Mission Planner, Solidworks, PrusaSlicer

May 2025 – February 2026

- Redesigned and optimized 4 structural points, reducing chassis weight from 650 g to 450 g while improving aerodynamic efficiency and mechanical robustness.
- Developed the electronic integration for a Pixhawk-based drone—covering GPS/IMU interfacing, telemetry links, and power routing—and executed extensive flight-testing in Mission Planner, optimizing PID tuning, thrust balance, vibration isolation, and GPS stability for reliable manual flight.

Martian Rover - Ganga | Altium, Teensy 4.1, Solidworks, PrusaSlicer

April 2025 – February 2026

- Engineered a robust system architecture with a simplified main PCB integrating dual Teensy 4.1 microcontrollers for full-rover control in IRC mission scenarios.
- Designed a custom Power Distribution Unit featuring dual 19V boost converters, dual bucks outputting 5V and 8V, 12V short-circuit protection, and a 5-channel solid-state relay to ensure reliable power delivery during field operations.
- Led the design, optimization, and validation of 25+ structural components throughout rover manufacturing, improving mechanical performance while adhering to the IRC weight limit of 65 kg.

VocalEyes | Python, Raspberry Pi 5, Solidworks, PrusaSlicer

February 2025 – December 2026

- Developed “VocalEyes,” a Raspberry Pi-based assistive device integrating I2S DAC Module and Pi Camera 3 Module with modular integration, achieving reliable real-time object awareness in diverse environments and providing low-cost accessibility to the visually impaired.
- Designed and 3D printed a self-ideated universal cap mount for Raspberry Pi 5 and other accessories, ensuring ergonomic usability and adaptability for users of any head size.

Martian Rover - Alaknanda | C/C++, Teensy 4.1, Solidworks, PrusaSlicer

August 2024 – February 2025

- Designed and implemented the electronics subsystem, integrating 6 sensors, 18 motor drivers, and power management circuits to ensure stable operation in diverse terrains.
- Led and mentored team in design and 3D printing of 45+ structural components using PrusaSlicer, significantly optimizing the rover’s mechanical stability and reducing overall weight.

DIY Water Meter | EasyEDA, Arduino IDE, ESP32, C++

June 2022 – September 2022

- Created a system to accurately measure water parameters, achieving consistent readings across 5 environmental conditions.
- Integrated advanced API functionalities into IoT systems to facilitate remote readings from the DIY water meter project, achieving precise measurements across 5 different parameters.

Publications in International and National Conferences

International Conference on Smart Cyber Physical Systems (ICSCPS, 2026) | Published

January 2026

- Contributed to the research paper by designing the system architecture of an ESP32-based outdoor multi-sensor tracker, integrating GNSS, thermal and RGB sensing, power management, and IoT data transmission into a unified embedded platform under the guidance of Dr. Amanpreet Kaur.

- Contributed to research on 4 custom PCB design, thermal management, and battery modeling for a multi-subsystem Martian rover, presented at a leading robotics conference under the mentorship of Dr. Ashish Singla and Dr. Sachin Kansal.

- Presented an extended abstract on a custom Martian-rover power system achieving 18 Nm torque and 86.7% efficiency while reducing heat-induced losses, guided by Dr. Ashish Singla and Dr. Sachin Kansal.

Leadership/Extracurricular

Overall Student Coordinator

July 2025 – July 2026*Mechatronics and Robotics Society, Thapar Institute of Engineering and Technology**Patiala, Punjab*

- Directed 2 teams comprising 60+ students under the guidance of Dr. Ashish Singla (President) and Dr. Sachin Kansal (Vice-President) for the International Rover Challenge, overseeing electronics integration, manufacturing, and 3D printing tasks, and led a separate team for the International Space Drone Challenge, managing overall drone development.
- Supervised the complete development of 2 ELC Projects integrating 50 electronic components for MARVIN - Cleaning Bot and Reconfig - Reconfigurable Bot, ensuring efficient system design and testing.
- Contributed to the design and integration of all systems in 3 minor projects under the MARS Society, improving prototyping efficiency and overall system reliability.

Lead - Electronics Department

June 2024 – June 2025*Mechatronics and Robotics Society, Thapar Institute of Engineering and Technology**Patiala, Punjab*

- Directed a team of 40+ students under the guidance of Dr. Ashish Singla (President) and Dr. Sachin Kansal (Vice-President) for the International Rover Challenge, managing electronics and 3D printing tasks.
- Supervised the electronics development of an ELC Project integrating 42 electronic components for a surveillance rover designed for military applications.
- Contributed to the design and integration of electronics systems in 5 ongoing minor projects under the MARS Society, enhancing prototyping efficiency and system reliability.

Technical Skills

Frameworks/Tools: Altium, EasyEDA, AutoDesk Eagle, Solidworks, Arduino IDE, STM32Cube IDE, Xilinx Vivado, Mentor Graphics.

Core Skills Knowledge: Schematic Capture, Power Electronics, PCB Design, Analog Electronics, Microcontrollers, Embedded Hardware, Robotics.

Certifications

- Certificate of Participation at International Conference on Smart Cyber Physical Systems, 2026
- Certificate of Completion of Qualcomm Technologies' 5G Introductory Level Course
- Certificate of Completion of 21st Century Technologies Course
- Certificate of Completion of The Altium Education PCB Basic Design Course
- Certificate of Participation at Advances In Robotics (AIR) 2025 7th International Conference of The Robotics Society
- Certificate of Participation in the Workshop on "Dynamics of Robotics System"
- Certificate of Participation for HackOwasp 6.0 Hackathon.
- Certificate of Appreciation for successful completion of ELC activity of IOT BASED SYSTEMS.
- Certificate of Appreciation for successful completion of ELC activity of DC Power Supply.
- Certificate of Appreciation for Microsoft Entrepreneurship Competition.

Honors and Awards

- Awarded for Champion at International Rover Competition, 2026.
- Awarded for 1st Runner Up at International Space Drone Competition, 2026.
- Awarded for 1st Runner Up at International Rover Competition, 2025.
- Awarded a 1 lakh Academic Scholarship from Thapar Institute of Engineering and Technology.